Thesis for the Degree of Doctor of Philosophy

Silicon nitride-based integrated components for optical communication

Alexander Caut



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Photonics Laboratory
Department of Microtechnology and Nanoscience - MC2
Chalmers University of Technology
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Abstract

To follow the trend of the data traffic and to limit the size of the hyperscale data centers, communication solutions offering small footprint, low cost and low power consumption are needed. Optical interconnects used in data centers are mostly short reach (approximately 100 m) based on GaAs-based 850 nm vertical-cavity surface emitting lasers (VCSELs) and OM4 multimode fibers (MMF). However, with 1 km-long optical links, the use of VCSEL-MMF at 850 nm becomes challenging at high data rates (Tb/s) due to large modal dispersion and high propagation loss. Therefore, other cost-effective methods are needed to compensate these limits. Single mode GaAs-based VCSELs have been demonstrated at 1060 nm of wavelength, where the chromatic dispersion is lower, for optical links ranging between 300 m and 10 km. This solution could be a better alternative than InP-based distributed feedback laser sources at 1310 nm in terms of cost and energy dissipation. As the modulation bandwidth of GaAs-based single mode VCSELs is limited to around 30 GHz, reaching the capacity target then requires a wavelength division multiplexing scheme with parallel single-core fibers (SCFs) or even multi-core fibers (MCFs)

In this thesis we discuss different types of demultiplexers at 1060 nm of wavelength. The proposed designed demultiplexers are arrayed waveguide gratings (AWGs) and cascaded Mach-Zehnder interferometers (MZIs). These two technologies are compared in terms of transmission, bandwidth, crosstalk and footprint with the number of output channels. Grating couplers at 1060 and 850 nm for on-chip coupling are also studied. The goal is to couple the light coming from a single mode fiber or a VCSEL with the lowest possible loss and back reflection.

Keywords: arrayed waveguide gratings, Mach-Zehnder interferometers, insertion loss, crosstalk, grating couplers, power splitters, polarization, silicon photonics, silicon nitride.

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Publications

This thesis is based on the work contained in the following papers:

[A] **Alexander Caut**, Marcello Girardi, Victor Torres-Company, Anders Larsson, and Magnus Karlsson, "Channel Scalability of Silicon Nitride (De-)multiplexers for Optical Interconnects at 1 μm", in *Journal of Lightwave Technology (JLT)*, 2023.
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[B] Mehdi Jahed, **Alexander Caut**, Jeroen Goyvaerts, Marc Rensing, Magnus Karlsson, Anders Larsson, Gunther Roelkens, Roel Baets, and Peter O'Brien, "Angled Flip-Chip Integration of VCSELs on Silicon Photonic Integrated Circuits", in *Journal of Lightwave Technology (JLT)*, Vol. 40, No. 15, pp. 5190-5200, August 1, 2022.
DOI: 10.1109/JLT.2022.3172781

[C] **Alexander Caut**, Vijay Shekhawat, Victor Torres-Company, and Magnus Karlsson, "Polarization-insensitive silicon nitride photonic receiver at 1 μm for optical interconnects", *Submitted for publication, 2023*

[D] **Alexander Caut**, Victor Torres-Company, and Magnus Karlsson, "Receiver platform based on thin Silicon Nitride waveguides at 1 μm", *Submitted for publication, 2023*

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[E] Marcello Girardi, Òskar B. Helgason, **Alexander Caut**, Magnus Karlsson, Anders Larsson, and Victor Torres-Company, "Multilayer integration in silicon nitride: decoupling linear and nonlinear functionalities for ultralow loss photonic integrated systems", in *Optics Express*, Vol. 31, No. 19, 11 Sep 2023.

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Other publications by the author, not included in this thesis, are:

[F] A. S. Alam, M. Girardi, **A. Caut**, A. Larsson, V. Torres-Company, M. Galili, Y. Ding, and K. Yvind, "LiNbO3/Si3N4-Bilayer Vertical Coupler for Integrated Photonics", *CLEO 2020*

[G] Marcello Girardi, Òskar B. Helgason, **Alexander Caut**, Magnus Karlsson, Anders Larsson, and Victor Torres-Company, "3D Integration of Microcombs", *CLEO 2023*

[H] Hezhi Zhang, Ching-Wen Shih, Denis Martin, **Alexander Caut**, Jean-François Carlin, Raphaël Butté and Nicolas Grandjean, "Short cavity InGaN-based laser diodes with cavity length below 300 μm", in *Semiconductor Science and Technology*, 34, 2019

[I] Hezhi Zhang, Ching-Wen Shih, Denis Martin, **Alexander Caut**, Jean-François Carlin, Raphaël Butté, and Nicolas Grandjean, "Broadened Bandwidth Amplified Spontaneous Emission from Blue GaN-Based Short-Cavity Superluminescent Light-Emitting Diodes", in *ECS Journal of Solid State and Technology*, 2020, 9 015019

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Acronyms

AWG	arrayed waveguide grating
MZI	Mach-Zehnder interferometer
GC	grating coupler
SMF	single-mode fiber
MMF	multi-mode fiber
CWDM	coarse wavelength-division multiplexing
DWDM	dense wavelength-division multiplexing
SNR	signal-to-noise ratio
BER	bit error rate

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CHAPTER 1

Introduction

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1.1 Network traffic

Global data traffic has been increasing at an exponential rate in the past decade. Mobile data traffic alone was 62 exabytes per month in 2020 and is expected to reach 607 exabytes in 2025 and 5016 in 2030 [1, 2]. While 5G technology is being released worldwide, 6G is already under development to meet the high demand for high-speed internet [2] and the backbone of the internet infrastructure are optical cables and data centers. Data centers consist of a vast amount of servers where most of the world's data is stored and accessed. These servers require a substantial amount of power and cooling. According to an European Commission report, data centers consumed 2.7 % of electricity within the EU in 2018 and could reach 3.21 % in 2030 [3]. This is due to the growing size and number of data centers. The number of hyperscale data centers was 259 in 2015, 448 in 2018 and reached 700 at the end of 2021 [4]. The largest data center measures now 1 000 000 m^2 [5]. This means that to limit the increasing size of the hyperscale datacenters, interconnect solutions offering small footprint, low cost and low power consumption will be needed. Today's optical interconnects used in data centers are short reach and use standard GaAs-based 850 nm vertical-cavity surface emitting lasers (VCSELs) and OM4 multimode fibers (MMFs) [6–8]. At this wavelength, the fiber links typically reach distances of 100 m [9] at

data rates of 25 Gb/s on-off keying (OOK) and 50 Gb/s pulse amplitude modulation 4-level (4PAM) [8, 10, 11]. 4PAM modulation is of interest as the data rate is increased by a factor of 2. However, the 4-PAM modulation format requires a higher signal-to-noise ratio (SNR) to get the same level of bit error rate (BER) than with OOK modulation [11].

1.2 VCSEL-based interconnects

1.2.1 Short-reach interconnects at 850 nm

GaAs-VCSELs at 850 nm are used in 95% of optical interconnects with typical lengths around 100 m [9, 12]. Data rates close to 100 Gb/s were achieved on these VCSELs with 4PAM [13]. Typical energy dissipations close to 100 fJ/bit at 50 Gb/s are reported in Ref [14]. However, the length of the optical links are limited due to chromatic dispersion in the fiber and propagation loss. The chromatic dispersion at 850 nm reaches -85 ps/nm/km in silica MMF and the propagation loss exceeds 2 dB/km [7, 15]. One possibility to reduce the chromatic dispersion effect consists in using single-mode (SM) VCSELs as the bandwidth is much narrower than for multimode ones [7]. Moreover, SM-VCSELs reduce the impact of modal dispersion if launched into carefully selected mode groups of the MMF [7,16]. Unfortunately, this also requires higher alignment precision as misalignment between SM-VCSELs and MMFs can lead to higher order modes excitation [7].

1.2.2 Medium-reach interconnects at 1060 nm

With the increasing size of hyperscale data centers and the limited length of short links at 850 nm, new solutions are needed to face this challenge. With the recent development of efficient GaAs SM-VCSELs emitting at 1060 nm [17] and with the reduced propagation loss and chromatic dispersion at 1060 nm (about 1 dB/km and -25 ps/nm/km respectively) [15], optical links at 1060 nm are of interest. After several efforts, with a pre-emphasis on GaAs-VCSELs at 50 Gb/s an error-free VCSEL-SMF 1-km long optical link was achieved without forward error correction (FEC) [18]. The used SMF was a prototype with negative dispersion. Links using InP distributed feedback lasers at 1310 nm are another alternative due to low propagation loss and chromatic dispersion but less efficient [8].

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1.2.3 Space division multiplexing (SDM)

To enable high data rate transmission, one existing technology based on multicore fibers (MCFs) consists of transmitting the data from an array of VCSELs, spatially matching the different cores of the fiber. According to Ref [19], the record demonstrated in SDM systems in 2020 was 10000 Tb/s. However, the spectral efficiency of this type of fiber is limited by the core count due to crosstalk coming from adjacent cores [19].

1.2.4 Wavelength division multiplexing (WDM)

WDM is another technology to reach high data rates. The two main types of WDM are coarse WDM (CWDM) where the channel spacing is usually 20-25 nm [20, 21] and dense WDM (DWDM) where the channel spacing is below 1 nm [22]. DWDM is the preferred scheme for long-haul communications in the C band (1530-1565 nm) due to the limited bandwidth of erbium doped fiber amplifiers (EDFAs). With the reduced footprint of VCSELs (250 x 250 μm^2) [20], it is possible to combine signals coming from different VCSELs (emitting at different wavelengths) into one fiber with the help of a multiplexer (MUX). Schemes using shortwave WDM that multiplex four wavelengths (850, 880, 910 and 940 nm) in a multimode fiber OM5 have been demonstrated in Ref [23, 24] with 4PAM modulation and enable up to 400 Gb/s [25]. However, parallel discrete VCSELs increase the footprint of the transmitter where there is a need for higher bandwidth densities. One solution to reduce the footprint consists in manufacturing multi-wavelength VCSELs arrays as proposed in Ref [8].

1.2.5 Role of silicon photonics

The technologies mentioned above allow a significant increase of the data rate in optical links, but with the substantial lack of space in the optical interconnects, the need of small device footprint and high bandwidth density require low cost and integrated solutions for space and wavelength division multiplexing.

On the one hand, electronic integrated circuits (EICs) are currently widespread in telecommunication and silicon is one of the dominating materials in the electronic market due to its properties, abundance and compatibility with CMOS processes. On the other hand, photonic integrated circuits (PICs) are also of interest for CMOS compatible wafers. Silicon photonics is also well established in the market due to its low cost

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and its CMOS-compatibility. For example, IBM is developing low-cost hybrid silicon electronic-photonic chips for high bandwidth transmission (10 Tb/s) [26]. In addition, according to Ref [27], silicon photonics is playing a key role in future generations co-packaging technology. To enable significant tighter integration between optical links and electronic integrated circuits, integration of lasers and electronics is essential. However, due to the indirect bandgap of silicon, electron-hole radiative recombination that is necessary to create stimulated emission cannot to occur. This is why semiconductor lasers are based on direct bandgap materials, typically in the III-V semiconductor region [28, 29].

One cost-effective technique used to integrate light sources such as VCSELs in PICs is to flip-chip bond the laser over the PIC with an optimized tilt-angle into a surface coupler. Flip-chip integration of VCSELs have been demonstrated previously in Ref [30] and is also discussed in Paper [B]. The flip-chip is realized with deposition of a solder ball between the VCSEL and the PIC, also allowing electrical contacting. Moreover, integration of laser sources via direct hetero-epitaxial growth of III-V materials on silicon have been demonstrated [31] and co-packaging of electronic and PICs led to recent demonstrations of 1.6 Tb/s transceivers [32]. In addition, silicon photonics enables compact and low cost integration of WDM components while being compatible with complementary metal-oxide-semiconductor (CMOS) processes [33]. For example, the arrayed waveguide grating (AWG) is a well known component for WDM [34–36] and can be integrated in silicon [22], silicon nitride [20], silica [37] and other materials.

1.3 This thesis

This thesis will first discuss in Chapter 2 the different usable materials for the integrated optical waveguides, the simulation methods to estimate the insertion loss and also the different waveguide designs. The focus of this thesis is on design, characterization and analysis of different types of fiber-chip couplers at 850 nm and 1060 nm. Chapter 3 of the thesis will focus on the transmitter side of the WDM optical interconnect at 1060 nm. Paper [A] expands the results of Chapter 3 and proposes two types of (de-)multiplexers which are (i) the arrayed waveguide gratings (AWGs) and (ii) the cascaded Mach-Zehnder interferometers (MZIs). A deep analysis on robustness of these two technologies to fabrication deviations (waveguide dimension, refractive index change), crosstalk, inser-

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tion loss, signal-noise-ratio (SNR), device's footprint with the number of channels and sensitivity to temperature compared to the VCSEL's temperature shift is presented as well. Chapter 4 will focus on grating couplers and a photonic integrated circuit including a flip-chipped VCSEL and a photodiode at 850 nm is demonstrated on Paper [B]. Chapters 5 will then focus on the receiver of the optical interconnect at 1060 nm. Papers [C] and [D] expands the results of Chapter 5 on the receiver at 1060 nm and focus on polarization insensitivity. Paper [C] proposes a receiver based on a square polarization independent waveguide and Paper [D] a receiver based on a thin waveguide. Chapter 6 proposes a photonic integrated circuit design based on micro-comb frequency source for optical interconnects at 1550 nm. The explored demultiplexers in this chapters are AWGs and a demonstration of a photonic integrated circuit is shown in Paper [E]. Finally, Chapter 7 presents an outlook for future work on these optical interconnects and Chapter 8 summarizes the main results in the papers.

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CHAPTER 2

Waveguide theory and simulation methods

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Design of integrated photonic devices requires selection of the appropriate material and the waveguide geometry depending on the wavelength and the device requirements. Indeed, the waveguide material will have an impact on the mode confinement, the device's footprint and the waveguide dimensions. The waveguide is either sandwiched between cladding layers made of silica for good mode confinement or just deposited on top of a silica cladding layer. For good mode confinement, the core index must be larger than the cladding index. The most commonly used waveguide materials in silicon photonics are silicon, silicon nitride and silica, and these will be discussed below.

2.1 Waveguide materials

2.1.1 Silicon waveguides

Silicon (Si) is a material that is CMOS-compatible, which is important for semiconductor devices. Due to high refractive index contrast between silicon and silicon dioxide (SiO_2), the mode confinement in the waveguide is excellent, allowing ultra-short bending radius (a few micrometers) and thus small footprint. However this also leads to strong roughness of waveguide sidewalls and higher scattering loss [38]. Si-waveguides are mostly based on a silicon-on-insulator (SOI) platform. In addition, the Si material has low absorption in the wavelength range 1.1-8.5 μm, which

means this material cannot be used for optical interconnects with wavelengths shorter than 1.1 μm (for example 850 nm or even 1060 nm) [39]. Finally, Si waveguides have an effective index that is very sensitive to waveguide width variation. Indeed, multiplexers (AWGs, MZIs) made of Si waveguides show a shift of 1 nm in the transmission spectra for a waveguide width deviation of 1 nm [39]. As we want to make robust designs to any fabrication deviation, this material is therefore less suitable for our devices.

2.1.2 Silicon nitride waveguides

Silicon nitride (SiN) waveguides make a good alternative to waveguides made of silicon. Also being a CMOS-compatible material, the refractive index of Si_3N_4 being around 2.00 at 1050 nm of wavelength, the refractive index contrast between the SiO_2 cladding and the waveguide is thus much smaller. This decreases the sensitivity of scattering loss due to sidewall roughness. Silicon nitride can be deposited either in low-pressure chemical vapor deposition (LPCVD) at 700°C, in plasma enhanced chemical vapor deposition (PECVD) at less than 400°C [38,40] and also with sputtering. The LPCVD deposition method has the advantage of keeping control of the SiN stoichiometry and is most suited for the telecom band (1550 nm) [38], and also in the 1 μm band. Moreover, SiN can be used for wavelengths much shorter than 1.1 μm as opposed to Silicon, and varying the N/Si ratio can cover wavelengths close to 400 nm [39]. The SiN waveguide is deposited with LPCVD in Paper [A] and PECVD in Paper [B] (at 850 nm of wavelength). The drawback of SiN with respect to Si is that its lower refractive index leads to lower mode confinement, which prevents small bending radius in curved waveguides and increases the device's footprint [41]. However, this is largely compensated by the ultra-low propagation loss obtained in SiN waveguides, which is an important priority for our research. Therefore, SiN was selected for fabrication in the cleanroom and platform development.

2.1.3 Silica waveguides

Silica (SiO_2) is another interesting alternative to silicon for wavelengths shorter than 1.1 μm. Due to low index contrast between the core and the cladding (also made of silica), the waveguide dimensions can almost match the fiber's core, enabling ultra low fiber-chip coupling loss (0.1 dB) [39]. In addition, this material can provide low propagation loss

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